

RAISING OF RICE NURSERY

RICE is an important cash crop grown with its coarse and fine cultivars in kharif. Although resources such as gravitational canal system for irrigation and environmental conditions etc., favour paddy production, the average per acre yields is 25 maunds per acre which is two to three times less than the yield in other rice producing countries. The reasons are poor practices adopted by farmers.

The rice crop is mostly transplanted by hand under puddle and non-puddle conditions. In some areas, resourceful farmers also practice dry and wet seeding, along with system of rice intensification (SRI) which is getting popular.

However, the best yield is achieved when the crop is sown at optimum time with management practices of weeds, diseases, and predators. Under such conditions, growth and yield depend upon the year-to-year seasonal patterns of solar radiation and temperature.

In the traditional system, it is very important to raise rice nursery and transplant it at an appropriate time to get optimum yield. It is recommended to raise rice nursery before May end as nursery seedlings raised before or after this period and then their early or late transplantation is prone to more bacterial blight and bakanea disease and increased attack by insect pests. Beside, high temperature at the stage of grain filling may produce empty spikelets. Because of the delay the kernel produced gets broken during threshing of harvested paddy. The delayed nursery transplanted seedlings produce low and weak tillering reducing final yield.

Nursery raising methods:

The main reason for raising a nursery is to provide seedlings without weeds. Transplantation of healthy, vigorous nursery seedlings without insect pests gives better yield. Method for raising nursery depends on local cultural practices, soil type and water availability, however, each method is harvested with uniform yield provided the healthy seedlings are transplanted. Nursery in Punjab is cultivated by three methods wet, dry and recently developed method using primed seeds.

Wet method is practiced in clayey and clay loam soils where standing water conditions can be created. This method is popular in traditional areas of rice cultivation. In this method, the selected field is filled with water 25-30 days before sowing nursery. However, it is better to plow the field twice in dry or watery conditions to break the soil clods in order to create better puddle conditions. After watering, the field is plowed twice followed by planking. The

same practice is repeated after 7-10 days and this whole operation is practiced depending on the water availability. After planking, the field is left for the weeds to grow and destroyed. After seed bed preparation, the field is divided into two plots by making bunds in the mid field and then sub-water channels are prepared in each half and three ridges are made at perpendicular to each channel dividing it into eight sub-plots, it will facilitate seed broadcast, irrigation and drainage practices. Then sprouted seeds are broadcasted in these plots. This method seems to be difficult and costly but allows less weed growth and seedlings raised thus are healthy and vigorous and become ready for transplanting within **25-30 days**.

For this nursery method, if farmers use their own seed stored at farm then it is better to use clean seed. Clean seed is placed between the water soaked layers of gunny bags for 24 hours under shade and also covered with gunny bags. To avoid damage by heat caused by suffocation, seed is turned with hands for proper aeration three times a day. After about 36-48 hours the sprouted seed is ready for sowing.

Then in puddle plots for nursery sowing, broadcast the **1 kg per marla** sprouted seeds of **coarse rice cultivars** IR-6 and KSK-282 while for **fine (Basmati) rice cultivars** use **0.5 -0.75 kg** seeds for raising nursery. At broadcast the water depth should be maintained at 1-1, 1/2 inch and don't drain out water at that day. Next day, drain out the water in the evening and irrigate in the morning and continue this practice for one week in hot sunny days. Gradually, check water drainage and increase the water level not more than three inches with the increasing seedling age.

If nursery seedlings are week, apply $\frac{1}{2}$ kg ammonium sulphate or urea $\frac{1}{4}$ kg per marla after 15-20 days of nursery sowing, if fertilizer applied to nursery is delayed death of seedling at the transplantation stage may take place. If more area is to be cultivated, it is better to use 30 days old seedlings to complete transplantation within a week.

The dry bed method is practiced in dry soil conditions and fields are prepared under dry conditions. Seeds beds of convenient dimensions are prepared by raising the soil to a height of about 5-10 cm. a thin layer of farmyard manure or half burnt paddy husk could be spread on the nursery bed mainly to facilitate uprooting. In this method, soaked seed are spread over the seedbed and then irrigated.

Rab method is practiced in D.G.Khan where soils are hard and uprooting of seedlings difficult. Nursery plots are leveled with a spade. Crop residues are spread uniformly on them and burnt a day before sowing. Seed rate is twice that of the wet bed method. Weed control is satisfactory. Seedlings are ready for transplanting in 35-40 days.

Another improved method is by using **primed seeds with CaCl_2 and KCl salts** solutions. It is being practiced by farmers in Sialkot and Faisalabad districts. Nursery seedlings raised thus are healthy, vigorous and ready for transplantation within 25-30 days. Such nursery transplanted seedlings result in better growth and yield better kernel quality at harvest and is also resistant to diseases. However, there is need of more research in this area and seeks extension services to transfer this technology to poor farmers.

While farmers practicing **system of rice intensification** should plant eight to 12 days old seedlings, so that root system may grow well and give 30 to 50 +tillers and result in high yields. For nursery raising healthy seeds at the rate of two kg/ac should be used. The pre-sprouted seeds should be sown on raised nursery bed. A layer of FYM can also be applied and sprouted seeds are sparsely spread on it and then again covered with another layer of manure. Nursery is properly and carefully irrigated and after 10-12 days seedlings are uprooted and transported to the field carefully.

Seed Dressing:

To control seed-born diseases such as bakaneas and brown leaf spot an easy and economical way is treatment of seed with proper fungicide. Seed treatment method depends on nursery raising method. Two methods used for seed treatment i.e. slurry and wet/solution method. In case of slurry method, fungicide is applied two weeks before sowing nursery. While in case of wet method seed is dipped in water.

Nursery pest management:

The nursery is also affected by different insects including stem borers, rice leaf folder and hispa, grasshopper and white backed plant hopper. Each insect attacks nursery at particular stage of its lifecycle to a threshold level. There is need of integrated approach to control these before their threshold level. Nursery raising using improved method is not a difficult task; however, it necessitates and depends on constant supervision.

Since transplanted rice requires high water input, labour and also more nutrient losses occurs under flooded conditions. Under the acute water shortage in the country due to increased urbanization, competition from non-agricultural sources and even non-availability of farm labour at critical times, there is need to focus towards water saving technologies in rice like direct seeded rice, alternate wetting and drying (AWD), system of rice intensification and transplanted rice under aerobic soil conditions to get benefit the resource poor farmers.